R You Ready? Cracking the Code of Employee Attrition

When I first started exploring why employees leave their jobs, I was struck by a surprising fact: many people often think age plays a huge role in retention. However, my recent project with IBM's employee data turned this idea on its head. My findings not only challenged my initial assumptions but also deepened my understanding of the complexities behind employee retention.

Why THIS Project?

My curiosity about employee retention began during my studies in psychology, where I learned about the relationships between various factors and behavior. This project allowed me to combine my interests in psychology and business, making it particularly meaningful to me. It was a chance to analyze real data and uncover insights that could help organizations better understand their workforce.

What Readers Will Gain

In this article, you'll learn about the surprising connections (or lack thereof) between age, income, and employee attrition. I'll share key findings from my analysis of IBM's HR data and offer insights that could help businesses improve their retention strategies.

Key Takeaways

- Age has a weak correlation (0.024) with hourly rates.
- Most correlations between attributes are positive but weak.
- Younger employees tend to leave the company more frequently.
- Total working years are a stronger predictor of monthly income than age.

Dataset Details

For this project, I used a dataset from IBM's human resource department, which consisted of 1,470 employees and 35 different attributes. This rich dataset provided a solid foundation for my analysis, allowing me to explore various factors related to employee retention.

Analysis Process

My analysis involved several steps:

1. **Data Cleaning**: I ensured the dataset was clean and ready for analysis.

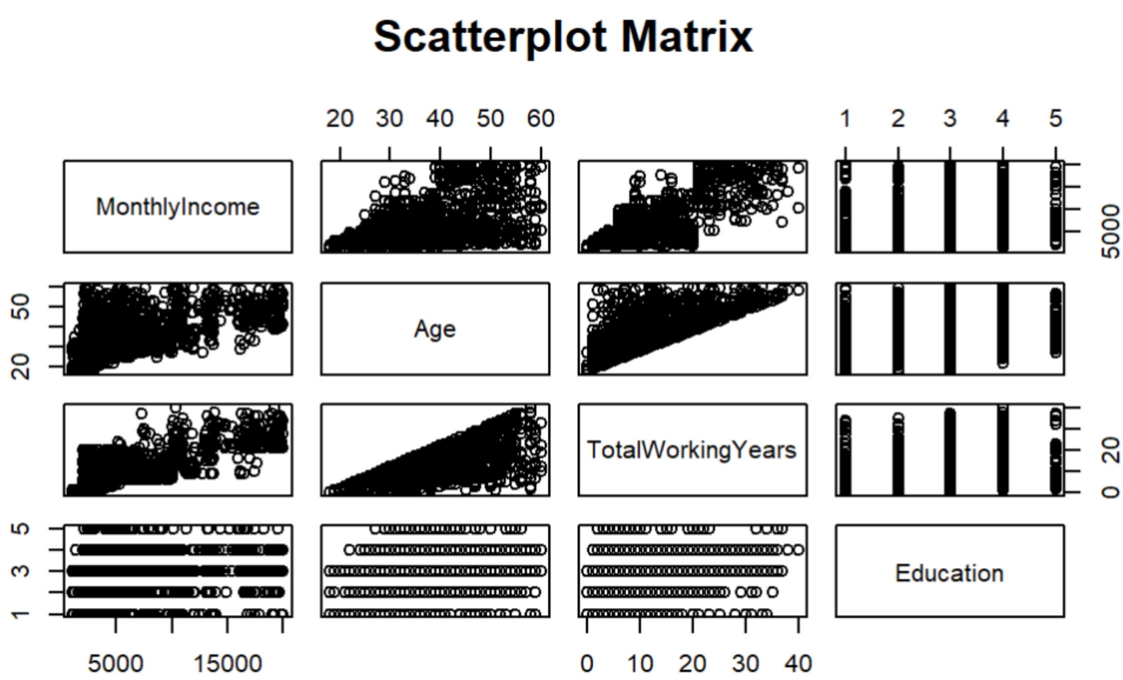
2. **Correlation Analysis**: I examined the relationships between demographics and attrition rates.
3. **Welch's Two Sample Test**: I compared groups of employees who left and those who stayed.
4. **Linear Regression**: I assessed how age and total working years predicted monthly income.

One of the surprising outcomes was the weak correlation I found between age and hourly rates. While I expected age to be a significant factor, it turned out to be quite minimal. It made me think about how many people might overlook other influential factors.

Visuals and Insights

Here are some visuals that illustrate my findings:

1. **Correlation Matrix**: This matrix displays the weak positive correlations among various demographics. Although all correlations are positive, they remain low, indicating that age isn't a strong predictor of retention.



	MonthlyRate	NumCompaniesWorked
Age	0.028051167	0.29963476
DailyRate	-0.032181602	0.03815343
DistanceFromHome	0.027472864	-0.02925080
Education	-0.026084197	0.12631656
HourlyRate	-0.015296750	0.02215688
MonthlyIncome	0.034813626	0.14951522
MonthlyRate	1.000000000	0.01752135
NumCompaniesWorked	0.017521353	1.00000000
TotalWorkingYears	0.026442471	0.23763859
TrainingTimesLastYear	0.001466881	-0.06605407

	TotalWorkingYears	TrainingTimesLastYear
Age	0.680380536	-0.019620819
DailyRate	0.014514739	0.002452543
DistanceFromHome	0.004628426	-0.036942234
Education	0.148279697	-0.025100241
HourlyRate	-0.002333682	-0.008547685
MonthlyIncome	0.772893246	-0.021736277
MonthlyRate	0.026442471	0.001466881
NumCompaniesWorked	0.237638590	-0.066054072
TotalWorkingYears	1.000000000	-0.035661571
TrainingTimesLastYear	-0.035661571	1.000000000

2. **Boxplot of Age vs. Attrition**: The boxplot clearly shows that younger employees are more likely to leave. The mean age of employees who left was 33.61, compared to 37.56 for those who stayed, highlighting the trend of younger employees facing higher attrition.


```
        Welch Two Sample t-test

data:  yes_age and no_age
t = -5.828, df = 316.93, p-value = 1.38e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -5.288346 -2.618930
sample estimates:
mean of x mean of y 
 33.60759  37.56123 
```

3. **Boxplot of Employee Number vs. Attrition**: This plot reveals that employee number doesn't significantly affect attrition rates, showing no meaningful difference between the groups.

4. **Linear Regression Comparison**: The graph illustrates that while age alone explains only 24% of the variance in monthly income, incorporating total working years boosts that to 60%, making it a much stronger predictor.

```
Call:
lm(formula = MonthlyIncome ~ Age, data = hrdata)

Residuals:
    Min      1Q  Median      3Q     Max 
-9990.1 -2592.7  -677.9  1810.5 12540.8 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -2970.67     443.70  -6.695 3.06e-11 ***
Age           256.57      11.67  21.995  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4084 on 1468 degrees of freedom
Multiple R-squared:  0.2479,	Adjusted R-squared:  0.2473 
F-statistic: 483.8 on 1 and 1468 DF,  p-value: < 2.2e-16
```

```
Call:
lm(formula = MonthlyIncome ~ Age + TotalWorkingYears, data =
hrdata)

Residuals:
     Min       1Q   Median       3Q      Max
-11310.8  -1690.8    -91.4   1428.3  11461.5

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        1978.08     352.36   5.614 2.36e-08 ***
Age                 -26.87      11.63  -2.311    0.021 *
TotalWorkingYears   489.13      13.65  35.824  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2984 on 1467 degrees of freedom
Multiple R-squared:  0.5988,    Adjusted R-squared:  0.5983
F-statistic:  1095 on 2 and 1467 DF,  p-value: < 2.2e-16
```

Main Takeaways

From this project, I learned that employee retention is influenced by a variety of factors, but age is not as significant as one might think. The analysis reinforced the idea that understanding the employee experience requires looking beyond traditional metrics. Organizations can benefit from focusing more on total working years, as it provides a clearer picture of income potential.

Conclusion and Personal Reflections

Reflecting on this project, I faced challenges, especially when interpreting the data and results. However, the surprises I encountered—like the minimal impact of age on income—helped reshape my perspective on employee retention. This experience has sparked a desire to continue exploring this field and contribute positively to businesses by leveraging data insights.

Call To Action

I invite you to connect with me on LinkedIn if you’re interested in discussing data analytics further or if you know someone looking to hire a data analyst. I’d love to hear your thoughts or questions on this topic!